

COMPUTING REGULAR SET TOLERANCES FOR COMBINATORIAL MINIMIZATION PROBLEMS

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Combinatorial minimization problems

A **combinatorial minimization problem** $\mathcal{P}$ is given by a tuple $(\mathcal{E}, D, c, f_c)$:

- $\mathcal{E}$ is a finite ground set of elements
- $D \subseteq 2^{\mathcal{E}} \setminus \{\emptyset\}$ is the set of feasible solutions
- $c : \mathcal{E} \rightarrow \mathbb{R}$ is the cost function
- $f_c : D \rightarrow \mathbb{R}$ is the objective (cost) function, which assigns costs to each feasible solution D

During the talk we will only consider objective function of type Σ :

$$\forall S \in D : f_c(S) = \sum_{e \in S} c(e).$$

A subset $S^* \subseteq \mathcal{E}$ is called an **optimal solution** of $\mathcal{P}$ if S^* is a feasible solution and the cost $f_c(S^*) = c^*$ of S^* is minimum:

$$S^* \in D \text{ and } f_c(S^*) = \min\{f_c(S) \mid S \in D\}.$$

Helpful notation

For $M \subseteq D$, we denote the cost of the best solution included in M by $f_c(M)$.

We denote the set of feasible solutions of D such that each of them does not contain $E \subseteq \mathcal{E}$ by $D_-(E)$:

$$D_-(E) = \{S \in D \mid E \cap S = \emptyset\}.$$

Analogously, we denote the set of feasible solutions of D such that each of them contains $E \subseteq \mathcal{E}$ by $D_+(E)$:

$$D_+(E) = \{S \in D \mid E \subseteq S\}.$$

We also denote the set of feasible solutions of D such that each of them does not contain $E \subseteq \mathcal{E}$ and contains $E' \subseteq \mathcal{E}$ by $D_\sim(E; E')$:

$$D_\sim(E; E') = \{S \in D \mid E \cap S = \emptyset, E' \subseteq S\}.$$

Single upper tolerance

Consider $e \in \mathcal{E}$. The **single upper tolerance** $u'_{\mathcal{P}}(e)$ is defined as the maximum by which the cost of e can be increased such that all optimal solutions S^* remain optimal. Note that the costs of all other elements $\bar{e} \in \mathcal{E} \setminus \{e\}$ must remain unchanged.

Proposition 1

If e belongs to some optimal solution S^* , then $u'_{\mathcal{P}}(e) = f_c(D_-(e)) - c^*$. Otherwise $u'_{\mathcal{P}}(e) = \infty$.

[GT25] G. Jäger and M. Turkensteen, Extending the definition of set tolerances, arXiv:2402.14542, 2025, <https://arxiv.org/abs/2402.14542>.

Consider $E = \{e_1, e_2, \dots, e_k\} \subseteq \mathcal{E}$. The (regular) set upper tolerance $u'_{\mathcal{P}}(E)$ of E is defined as the maximum of all those α such that the costs of all elements $e \in E$ are not decreased, the sum of all increases equals α , and all optimal solutions S^* remain optimal. Note that the costs of all elements $\bar{e} \in \mathcal{E} \setminus E$ must remain unchanged.

Single lower tolerance

Consider $e \in \mathcal{E}$. The **single lower tolerance** $l'_{\mathcal{P}}(e)$ is defined as the maximum by which the cost of e can be decreased such that the objective value of optimal solution remains the same. Note that the costs of all other elements $\bar{e} \in \mathcal{E} \setminus \{e\}$ must remain unchanged.

Proposition 2

$$l'_{\mathcal{P}}(e) = f_c(D_+(e)) - c^*.$$

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Consider $E = \{e_1, e_2, \dots, e_k\} \subseteq \mathcal{E}$. The (regular) set lower tolerance $u'_{\mathcal{P}}(E)$ of E is defined as the maximum of all those α such that the costs of all elements $e \in E$ are not increased, the sum of all decreases equals α , and the objective value of optimal solution remains the same. Note that the costs of all elements $\bar{e} \in \mathcal{E} \setminus E$ must remain unchanged.

Computing set upper tolerance

Consider a set $E = \{e_1, e_2, \dots, e_k\} \subseteq \mathcal{E}$. Let S^* be an optimal solution of $\mathcal{P}$ with optimal objective value c^* and let $E' = E \cap S^*$. Also let $\alpha_1, \alpha_2, \dots, \alpha_k \geq 0$ be the increases in the costs of the elements $e_1, e_2, \dots, e_k$, respectively.

Computing set upper tolerance

We introduce the following linear program called “Exact Upper LP” (EUL):

$$\begin{aligned} & \text{maximize} && \sum_{i=1}^k \alpha_i \\ & \text{subject to} && \\ & && \sum_{e_i \in E'} \alpha_i = \sum_{e_i \in F} \alpha_i \quad \forall F \subseteq E \text{ with } f_c(D_{\sim}(E \setminus F; F)) = c^* \\ & && c^* + \sum_{e_i \in E'} \alpha_i \leq f_c(D_{\sim}(E \setminus F; F)) + \sum_{e_i \in F} \alpha_i \\ & && \quad \forall F \subseteq E \text{ with } f_c(D_{\sim}(E \setminus F; F)) > c^* \\ & && \alpha_i \geq 0 \quad \forall i \in [k] \end{aligned}$$

Computing set upper tolerance

Theorem 1

The optimal objective value of EUL is equal to the set upper tolerance $u'_{\mathcal{P}}(E)$.

Computing set lower tolerance

Consider a set $E = \{e_1, e_2, \dots, e_k\} \subseteq \mathcal{E}$. Let c^* be the optimal objective value of $\mathcal{P}$. Also let $\alpha_1, \alpha_2, \dots, \alpha_k$ be the decreases in the costs of elements $e_1, e_2, \dots, e_k$, respectively.

Next we will introduce three linear programs.

Computing set lower tolerance

“Exact Lower LP” (ELL):

$$\begin{array}{ll} \text{maximize} & \sum_{i=1}^k \alpha_i \\ \text{subject to} & \\ & \sum_{e_i \in F} \alpha_i \leq f_c(D_{\sim}(E \setminus F; F)) - c^{\star} \quad \forall F \subseteq E \\ & \alpha_i \geq 0 \quad \forall i \in [k] \end{array}$$

Computing set lower tolerance

“Include Lower LP” (ILL):

$$\begin{array}{ll} \text{maximize} & \sum_{i=1}^k \alpha_i \\ \text{subject to} & \\ & \sum_{e_i \in F} \alpha_i \leq f_c(D_+(F)) - c^\star \quad \forall F \subseteq E \\ & \alpha_i \geq 0 \quad \forall i \in [k] \end{array}$$

Computing set lower tolerance

“Tolerance Lower LP” (TLL):

$$\text{maximize} \quad \sum_{i=1}^k \alpha_i$$

subject to

$$\sum_{i=1}^k \alpha_i \leq f_c(D_+(E)) - c^\star$$

$$\sum_{e_i \in F} \alpha_i \leq l'_{\mathcal{P}}(F) \quad \forall F \subsetneq E, F \neq \emptyset$$

$$\alpha_i \geq 0 \quad \forall i \in [k]$$

Theorem 2

The optimal objective values of ELL, ILL and TLL are equal to the set lower tolerance $l'_{\mathcal{P}}(E)$.

Computing set lower tolerance for sets of small cardinality

Proposition 3

Let $E = \{e_1, e_2\}$. Then $l'_{\mathcal{P}}(E) = \min \{f_c(D_+(E)) - c^*, l'_{\mathcal{P}}(e_1) + l'_{\mathcal{P}}(e_2)\}$.

Proposition 4

Let $E = \{e_1, e_2, e_3\}$. Then

$$l'_{\mathcal{P}}(E) = \min \left\{ \begin{array}{l} f_c(D_+(E)) - c^*, l'_{\mathcal{P}}(e_1) + l'_{\mathcal{P}}(e_2, e_3), \\ l'_{\mathcal{P}}(e_2) + l'_{\mathcal{P}}(e_1, e_3), l'_{\mathcal{P}}(e_3) + l'_{\mathcal{P}}(e_1, e_2), \\ \frac{1}{2} \cdot (l'_{\mathcal{P}}(e_1, e_2) + l'_{\mathcal{P}}(e_1, e_3) + l'_{\mathcal{P}}(e_2, e_3)) \end{array} \right\}.$$

Computing set upper tolerance for sets of small cardinality

Proposition 5

Let $E = \{e_1, e_2\} \subseteq \mathcal{E}$. The following statements hold:

(a) Let $f_c(D_+(E)) = c^*$ hold. Then it holds:

$$\begin{aligned} u'_{\mathcal{P}}(E) &= \min \{ f_c(D_{\sim}(e_1; e_2)) - c^* + f_c(D_{\sim}(e_2; e_1)) - c^*, f_c(D_-(E)) - c^* \} \\ &= \min \{ f_c(D_-(E)) - c^*, u'_{\mathcal{P}}(e_1) + u'_{\mathcal{P}}(e_2) \}. \end{aligned}$$

(b) Let **(a)** does not hold and $f_c(D_{\sim}(e_1; e_2)) = f_c(D_{\sim}(e_2; e_1)) = c^*$ hold. Then it holds

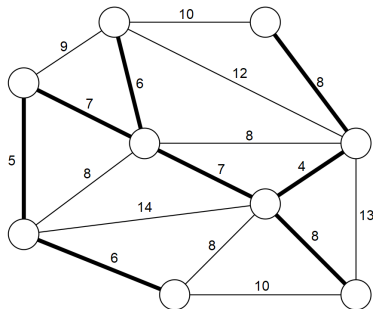
$$u'_{\mathcal{P}}(E) = 2 \cdot (f_c(D_-(E)) - c^*).$$

(c) Let **(a)** and **(b)** do not hold. Then $u'_{\mathcal{P}}(e) = \infty$ holds.

For cardinality 3 we have 6 different cases.

Minimal spanning tree problem

The minimum spanning tree problem (MSTP) is the problem of finding a spanning tree of minimum weight in a given weighted connected undirected graph.



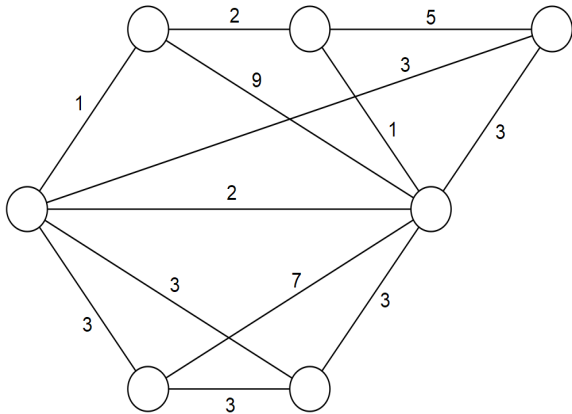
Minimal spanning tree problem

Formally, let Γ be a connected undirected graph with weight function $c : E(\Gamma) \rightarrow \mathbb{R}$. Then MSTP is a combinatorial minimization problem $\mathcal{P}$ given by a tuple $(E(\Gamma), D, c, f_c)$, where D is the set of all spanning trees in Γ and $f_c : D \rightarrow \mathbb{R}$ is the objective function of type $\sum$ defined by $f_c(T) = \sum_{e \in E(T)} c(e) \ \forall T \in D$.

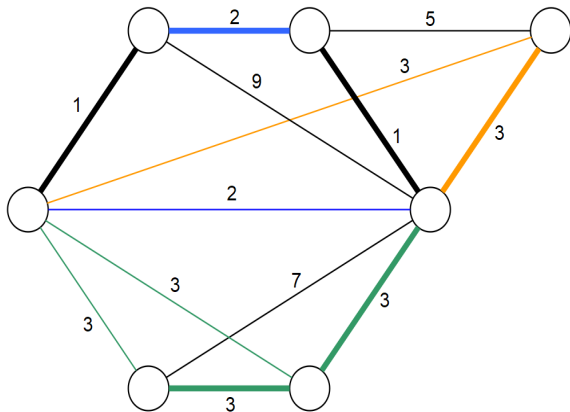
Theorem 3

For a set of edges $G = \{g_1, g_2, \dots, g_k\} \subseteq E(\Gamma)$ the lower set tolerance $l'_{\mathcal{P}}(G)$ equals $\sum_{i=1}^k l'_{\mathcal{P}}(g_i)$.

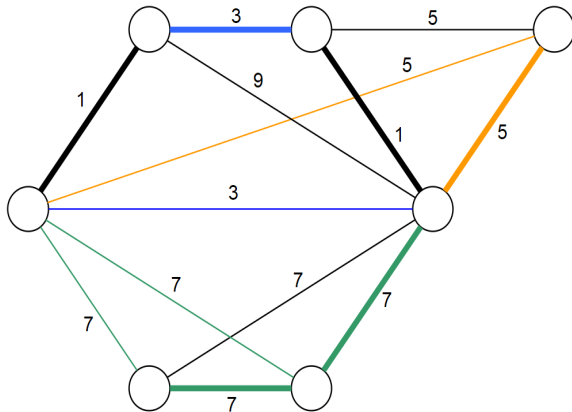
Upper set tolerance for MSTP



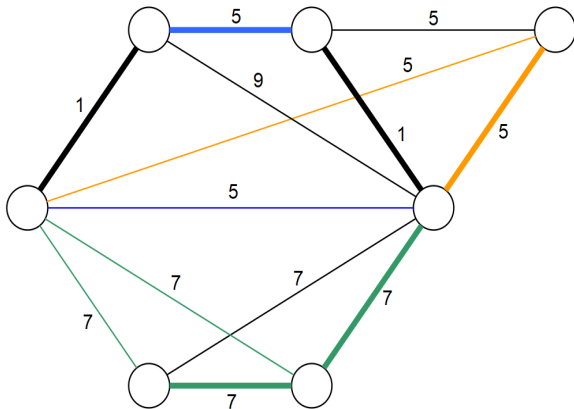
Upper set tolerance for MSTP



Upper set tolerance for MSTP



Upper set tolerance for MSTP



Thanks for your attention!